SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 2.2: Plant Transport — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 10	
Sub-Strand	Sub-Strand 2.2: Plant Transport
Driving Question	DRIVING QUESTION: How does water and food travel through an entire plant — from the soil to the highest leaf — and why does this journey sometimes fail, causing crops like maize in Kibwezi to wilt and die?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring Phenomenon — Wilting Maize in Kibwezi**	Students observed photographs and/or a short video of maize crops in Kibwezi wilting during a dry spell, despite green leaves and apparently healthy stems. They also observed a celery stalk (or white carnation) placed in coloured water and noted that colour travelled upward through the stem over time. Teacher should cold-call at least 4 students to describe what they see before offering any vocabulary — this protects initial observations from teacher bias.	Students learned that a living plant must continuously move materials — water, minerals, and sugars — from where those materials enter or are made to where they are needed. The wilting maize phenomenon establishes that this transport system can FAIL, and that failure has real consequences for food security in semi-arid regions like Kibwezi, Makueni County. Students also learned the meaning of 'anchoring phenomenon' as the central mystery the class will keep returning to across all 8 lessons.	Students explained (in initial, rough terms) that 'something must be broken or blocked' when maize wilts — the water that enters the roots is not reaching the leaves. Teacher should record all student ideas on the Driving Question Board (DQB) without correcting them at this stage; misconceptions are valuable diagnostic data. Pedagogical note: preserve student language exactly on the DQB — do NOT rephrase into scientific vocabulary yet. This baseline record will be revisited and revised in Lesson 8.
Lesson 2 **What Do Plants Need to Survive? — Mapping the Case for a Transport System**	Students examined diagrams and data cards showing where key plant resources enter the plant (water and minerals → roots; carbon dioxide → leaves; light → leaves) and where those resources are consumed (water used in photosynthesis at leaves; minerals used for growth at shoot tips; sugars produced at leaves but needed at roots). Students mapped these mismatches on a large class diagram of a maize plant, using arrows to show the 'resource gap' that transport must bridge.	Students learned that because plant resources enter at one location but are needed and used at a different location (e.g., water enters at the root but is needed at the leaf for photosynthesis), a specialised internal transport system is essential. Without transport, cells far from the resource entry point would starve or dehydrate. This is why a single-celled organism like Euglena does not need a transport system, but a tall maize plant in Kibwezi does.	Students explained the 'resource mismatch' problem using their annotated maize diagrams, linking it back to the driving question: if the transport system fails to bridge the gap between roots (water entry) and leaves (water use), the maize wilts. Teacher should use a think-pair-share with the prompt: 'Why can a tiny water plant like Elodea survive without transport tubes, but a maize plant cannot?' Allow 60 seconds of silent thinking before pairs discuss. Cold-call 3 pairs to share reasoning.

Lesson 3 **The Big Picture: Overview of the Plant Transport System (Xylem and Phloem)**	Students examined cross-section slides (or high-quality photomicrograph cards) of a maize stem, identifying two distinct tissue types within vascular bundles. They also watched a short animation showing water molecules moving upward through xylem vessels and sucrose moving bidirectionally through phloem. Teacher should pause the animation at key moments and ask: 'What direction is this molecule moving — and why might that direction matter for the Kibwezi maize?'	Students learned that plants have two specialised vascular transport tissues: (1) Xylem — transports water and dissolved mineral salts from roots upward to leaves; made of dead, hollow, lignified cells; and (2) Phloem — transports dissolved organic solutes (sucrose and amino acids) from leaves to all other parts of the plant; made of living sieve tube cells and companion cells. These two tissues together form the vascular bundle, visible in both stem and leaf.	Students explained the functional difference between xylem and phloem using a two-column comparison table and annotated the cross-section diagram with both tissue labels. They then linked this overview to the driving question: wilting in Kibwezi maize could involve failure in xylem (water not reaching leaves), phloem (sugars not reaching roots), or both. Pedagogical note: insist students use the word 'because' in every explanation — this forces causal reasoning rather than simple labelling.
Lesson 4 **The Entry Point: Root Structure and Water Absorption**	Students observed a demonstration or simulation of osmosis using visking tubing (or a raw potato strip) placed in solutions of different concentrations, recording changes in mass or volume over 15 minutes. They also examined diagrams of root hair cells, noting the large surface area created by the elongated hair-like projection. Teacher should ask: 'What does the shape of a root hair cell tell you about its function?' — wait 30 seconds before taking responses.	Students learned that water enters the root by osmosis — moving passively from soil water (high water potential) through root hair cells, across the cortex, through the endodermis, and into the xylem. Root hair cells are adapted for absorption by their large surface area, thin cell wall, and presence of a large central vacuole. The Casparian strip in the endodermis forces water through cell cytoplasm (symplast pathway), allowing selective control of mineral uptake into the xylem.	Students explained the step-by-step journey of a single water molecule from soil to xylem vessel, using a flow diagram. They connected this to the Kibwezi context: during drought, soil water potential drops — the osmotic gradient between soil and root hair cells decreases or even reverses, so water absorption by osmosis slows or stops, explaining why maize wilts even before all soil water is gone. Pedagogical note: many students confuse osmosis with active transport here — address this explicitly using the 'no energy needed' rule for osmosis.
Lesson 5 **The Highway: Stem Structure and Vascular Bundles**	Students dissected a fresh maize stem (or bean stem) cross-section and used a hand lens or low-power microscope to locate vascular bundles. They compared the arrangement of vascular bundles in a monocot stem (maize — scattered) versus a dicot stem (bean — ring arrangement), recording observations as labelled drawings. Teacher should circulate and ask each group: 'Which tissue is closer to the inside of the bundle — and what structural feature helps you tell xylem from phloem?'	The stem contains vascular bundles made up of two distinct tissues: xylem (inner, thick-walled, dead cells forming hollow tubes for carrying water and minerals upward) and phloem (outer, living sieve tubes for carrying dissolved sugars). Lignin in xylem cell walls provides mechanical support, helping the maize stem stand upright. In a monocot like maize, vascular bundles are scattered throughout the stem; in dicots like beans, they form a ring. The stem acts as the 'highway' connecting the absorbing roots to the photosynthesising leaves.	Students explained how the structural features of xylem vessels — dead cells, no cross-walls, lignified walls — make them ideal hollow pipes for carrying water under tension without collapsing. They linked stem vascular anatomy to the driving question: if xylem vessels become blocked (e.g., by air bubbles during drought, or by fungal pathogens like Fusarium common in Kenyan maize fields), water flow to leaves is interrupted and wilting follows. Pedagogical note: the 'dead cells = hollow tube' logic is a classic exam point — reinforce with at least one worked exam-style question in class.
Lesson 6 **The Destination: Leaf Structure and Transpiration**	Students conducted the cobalt chloride paper experiment (or Vaseline leaf experiment) to detect and compare rates of water loss from the upper and lower	Students learned that transpiration is the loss of water vapour from a plant, mainly through stomata located predominantly in the lower epidermis of leaves. The rate of	Students explained the adaptive trade-off of stomata: open stomata allow CO_2 in for photosynthesis but also allow water vapour to escape. During the dry season in

	surfaces of a maize or bean leaf. They recorded colour-change times as quantitative data and calculated which surface lost more water. Teacher should prompt: 'Before you start, predict — which surface will change the cobalt chloride paper faster, and why?' Record class predictions publicly before data collection begins.	transpiration is affected by temperature, humidity, wind speed, and light intensity — all factors highly variable in Kenya's semi-arid zones like Kibwezi. Guard cells control stomatal opening and closing: when turgid (water-filled), guard cells bow apart and open the stoma; when flaccid (water-deficient), they close, reducing water loss but also limiting CO_2 entry for photosynthesis.	Kibwezi, maize guard cells close stomata to conserve water, but this halts photosynthesis, starving the plant of sugars — a double failure. Students drew and annotated stomata in open and closed states, labelling guard cells, chloroplasts, and the stomatal pore. Pedagogical note: use a local weather data set from Kibwezi (Kenya Meteorological Department records) to make transpiration rate calculations authentic and relevant.
Lesson 7 **The Engine: Transpiration Pull, Cohesion-Tension, and Root Pressure**	Students observed a potometer experiment (or teacher-led demonstration using a leafy shoot and capillary tube) measuring the rate of water uptake under different conditions (fan on vs. fan off; light vs. dark; humid vs. dry). They plotted results as a bar chart and identified which variable caused the greatest change in uptake rate. Teacher should ask: 'If the shoot takes up more water when the fan is on, what does that tell us about what is PULLING the water up?'	Students learned that water moves up the xylem primarily through the cohesion-tension mechanism: (1) transpiration at the leaf surface removes water molecules, lowering water potential in leaf cells; (2) water is pulled up the xylem by tension (negative pressure) created by this evaporation; (3) water molecules stick together (cohesion) and to xylem walls (adhesion), forming an unbroken column from root to leaf. Root pressure (generated by active ion uptake creating osmotic pull) provides a supplementary push, especially at night. Together these forces explain how water can rise 30 m up a tall tree — or travel up a 2-metre maize stem in Kibwezi.	Students explained why breaking the water column — through drought-induced cavitation (air bubbles entering xylem when water is under extreme tension) — causes irreversible wilting in Kibwezi maize. Once the column breaks, transpiration pull cannot operate and water ceases to rise. Students constructed a force-diagram showing the relative contributions of transpiration pull vs. root pressure at different times of day. Pedagogical note: the concept of 'negative pressure pulling water' is deeply counterintuitive — use the analogy of a drinking straw to build intuition before introducing formal terminology.
Lesson 8 **The Return Journey: Translocation and Final Model Build**	Students examined aphid stylet experiments (diagrams/data) showing phloem sap composition and pressure, and reviewed autoradiograph images demonstrating that radioactively labelled sucrose moves from leaves to roots and to growing shoot tips. They also examined data showing that phloem transport is bidirectional — sucrose moves from any 'source' (leaf) to any 'sink' (root, fruit, young leaf). Teacher should ask: 'How is the direction of phloem transport different from xylem transport — and why does that difference matter?'	Students learned that phloem sieve tubes, assisted by companion cells, transport sucrose and amino acids from photosynthetic source cells (leaves) to all metabolically active or storage sink cells (roots, fruits, seeds, young leaves) by the pressure-flow (mass flow) mechanism. High sucrose concentration at the source lowers water potential, drawing water in from adjacent xylem and raising hydrostatic pressure; at the sink, sucrose is unloaded, water potential rises, water leaves phloem to xylem, and pressure drops — creating a pressure gradient that drives bulk flow from source to sink.	Students explained how, during drought in Kibwezi, reduced photosynthesis (because stomata close to conserve water) means less sucrose is loaded into phloem at source cells — so translocation slows, roots and developing maize cobs are starved of sugars, growth stops, and cobs fail to fill. Students then completed their FINAL cumulative model of the maize plant transport system, annotating the full water pathway (soil → root hair → cortex → xylem → stem → leaf mesophyll → stomata → atmosphere) and the full sucrose pathway (leaf mesophyll → phloem → stem → root/cob). Pedagogical note: final models should be compared with the rough initial models from Lesson 1 so students can explicitly articulate what they now understand that they did not before — this

			metacognitive step is a KICD CBE core competency requirement.